Premraj Prakare Python Test7

```python
class Book:
    def __init__(self, bid, name, price, author):
        self.bid = bid
        self.name = name
        self.price = price
        self.author = author

def add_book(books):
    bid = int(input("Enter book ID: "))
    name = input("Enter book name: ")
    price = float(input("Enter book price: "))
    author = input("Enter author name: ")
    books[bid] = Book(bid, name, price, author)
    print("Book added successfully!")

def delete_book(books):
    bid = int(input("Enter book ID to delete: "))
    if bid in books:
        del books[bid]
        print("Book deleted successfully!")
    else:
        print("Book not found")

def display_books(books):
```

```
    if not books:

        print("No books in the database.")

    else:

        for book in books.values():

            print(f"ID: {book.bid}, Name: {book.name}, Price:
{book.price}, Author: {book.author}")


def search_book(books):

    bid = int(input("Enter book ID to search: "))

    if bid in books:

        book = books[bid]

        print(f"ID: {book.bid}, Name: {book.name}, Price:
{book.book.price}, Author: {book.author}")

    else:

        print("Book not found")


def main():

    books = {} # Dictionary to store book objects


    while True:

        print("\nMenu:")

        print("a. Add Book")

        print("b. Delete book by id")

        print("c. Display all books")

        print("d. Search book by id")

        print("e. Exit")
```

```
        choice = input("Enter your choice: ").lower()


        if choice == "a":
            add_book(books)
        elif choice == "b":
            delete_book(books)
        elif choice == "c":
            display_books(books)
        elif choice == "d":
            search_book(books)
        elif choice == "e":
            break
        else:
            print("Invalid input")


if __name__ == "__main__":
    main()
output:
```

```
----------Library----------:
1. Add Book
2. Delete book by id
3. Display all books
4. Search book by id
0. Exit
Enter your choice: 1
Enter book ID: 1
Enter book name: chava
Enter book price: 2400
Enter author name: shivaji sawant
Book added successfully!

----------Library----------:
1. Add Book
2. Delete book by id
3. Display all books
4. Search book by id
0. Exit
Enter your choice: 3
ID: 1, Name: chava, Price: 2400.0, Author: shivaji sawant
```

```python
# q2
list1 = ["this","is","variable","list","comprehension"]
list2 = [ x.upper() for x in list1 ]
print("Original list :",list1)
print("Uppercase list :",list2)
```

```
PS D:\Project\Python2025> & C:/Users/91848/AppData/Local/Programs/Python/Python311/python.exe d:/Project/Python2025/Test7/t7_q.py
Original list : ['this', 'is', 'variable', 'list', 'comprehension']
Uppercase list : ['THIS', 'IS', 'VARIABLE', 'LIST', 'COMPREHENSION']
PS D:\Project\Python2025>
```

```python
# q3
list1 = [x for x in range(1,51) if x % 2 == 0]
print("Even numbers from 1 to 50:", list1)
```

output:

```
PS D:\Project\Python2025> & C:/Users/91848/AppData/Local/Programs/Python/Python311/python.exe d:/Project/Python2025/Test7/t7_q.py
Even numbers from 1 to 50: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50]
PS D:\Project\Python2025>
```